Latent Diffusion Prior-Enhanced Frequency-Aware Deep Unfolding Network for Hyperspectral Image Denoising

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***Abstract*—Deep unfolding networks (DUNs) offer an iterative paradigm that unrolls optimization procedures into a cascaded network structure for hyperspectral image (HSI) denoising. However, existing DUNs for HSI denoising suffer from two notable limitations: 1) the ill-posed inverse problem of handling severely degraded observations, calling for a more powerful degradation-free prior that encodes the spatial–spectral structure of HSIs and 2) existing discriminative denoisers are deterministic and tend to produce oversmoothed results. On the contrary, stochastic latent diffusion models (LDMs) have demonstrated the potential to synthesize high-frequency details with high perceptual quality for HSIs, but often at the cost of sacrificing data fidelity. To this end, we propose a latent diffusion prior-enhanced frequency-aware DUN (Diff-DUN) for HSI denoising. It builds on a half-quadratic splitting formulation that reconciles the observation consistency of deterministic DUNs with the high perceptual quality of a stochastic LDM. In particular, we employ a two-phase training strategy to first learn degradation-free priors from clean HSIs, followed by training the LDM to generate these priors conditioned on noisy inputs. Furthermore, we propose a frequency-aware Tetra Transformer (TT) block as the core component within the denoiser, introducing explicit frequency flow and cross-flow interactions to jointly model spatial, spectral, and frequency characteristics for more faithful recovery. Comprehensive evaluations across multiple datasets demonstrate that Diff-DUN achieves a favorable tradeoff between denoising accuracy and computational overhead, performing competitively with state-of-the-art (SOTA) methods.**

***Index Terms*—Deep unfolding networks (DUNs), frequency information, hyperspectral image (HSI) denoising, latent diffusion model (LDM).**

I. Introduction

HYPERSPECTRAL images (HSIs), with hundreds of narrow spectral bands, enable precise material identification in applications such as precision agriculture [1], classification [2], [3], and geological surveying [4]. Nevertheless, real-world HSIs are inevitably degraded by multiple noise types during acquisition. Such degradation not only disrupts spatial structures but also distorts the spectral signatures for material identification, thereby affecting subsequent analysis. Thus, an effective HSI denoising method must preserve both spatial details and spectral fidelity while suppressing noise.

Over the past decades, two main categories of methods have been proposed to address this issue, i.e., model-based methods [5], [6], [7], [8], [9], [10], [11] and deep learning (DL) methods [12], [13]. Recently, deep unfolding networks (DUNs) [14], [15], [16], [17] have shown remarkable effectiveness by integrating model-based optimization with deep priors via a staged architecture. At each stage, a data-fidelity term ensures observation consistency, while a regularization term encodes prior knowledge. Compared with traditional handcrafted regularizers, this paradigm not only accelerates iterative convergence but also captures richer priors with enhanced adaptability. However, existing DUNs for HSI denoising exhibit two critical limitations.

1) Existing methods suffer from the performance bottleneck caused by the intrinsic ill-posedness in handling severe degradation. This calls for a more powerful degradation-free prior that can faithfully encode spatial–spectral characteristics.
2) Most denoisers rely on deterministic mappings, which tend to produce oversmooth and conservative estimates under heavy noise. This behavior exemplifies the "regression to the mean" [18], revealing that deterministic regression alone is insufficient to preserve fine spatial–spectral structure.

In contrast to the above deterministic denoisers, stochastic denoising diffusion models have achieved considerable success in synthesizing HSIs [19] with high perceptual quality by exploring diverse plausible solutions. However, their iterative sampling process incurs high computational cost. To mitigate this issue, latent diffusion models (LDMs) [20] perform diffusion in a low-dimensional latent space, offering improved efficiency while retaining strong generative ability. Hence, leveraging such LDMs to conduct stochastic sampling for HSI denoising has attracted increasing attention [21], [22], [23]. Nevertheless, for LDMs, the emphasis on high perceptual quality often comes at the cost of uncertainty and hallucinations [24], [25], [26] that compromise observation consistency.

Received 13 August 2026; accepted 27 August 2026. Date of publication 7 September 2026; date of current version 22 September 2026. This work was supported in part by the NSFC Regional Innovation and Development Joint Fund under Grant U25A20404, in part by NSFC under Grant 62271418 and Grant 62301456, in part by the Postdoctoral Fellowship Program of China Postdoctoral Science Foundation under Grant 2025T180066 and Grant GZB20230605, in part by the State Key Laboratory of Intelligent Geotechnics and Tunneling under Grant SKL2024-B2, and in part by the Fundamental Research Funds for the Central Universities under Grant 2682026ZD005. *(Corresponding author: Heng-Chao Li.)*

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Digital Object Identifier 10.1109/TGRS.2026.3731221

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Therefore, it remains a critical challenge to efficiently reconcile the stochastic nature of LDMs with the observation consistency of deterministic mappings for high-quality HSI denoising. To tackle this issue, incorporating an LDM as a prior provider within a DUN presents a promising paradigm [27], [28], [29].

To this end, inspired by the emerging paradigm [29], we propose a diffusion prior-enhanced frequency-aware DUN (Diff-DUN) specifically reformulated for HSI denoising. We reformulate the optimization procedure into a deterministic half-quadratic splitting (HQS)-based framework, where the learned prior is integrated via a denoiser at each stage. Since the priors extracted from degraded HSIs alone are insufficient for severely corrupted spatial–spectral structures of HSIs, we follow the two-phase training strategy [29] to learn a degradation-free prior. Furthermore, to incorporate the prior, we propose the frequency-aware Tetra Transformer (TT) block as the core component of the U-shaped denoiser. The TT block integrates the degradation-free prior flow with spatial, spectral, and frequency flows through progressive interaction. In particular, the spatial and spectral flows, respectively, capture local spatial correlations and interband dependencies to counteract noise, and jointly construct an observation feature. The feature is then employed to adaptively select observation-consistent information from the degradation-free prior, thereby preventing this prior from weakening observation consistency. Meanwhile, the explicit frequency flow disentangles the frequency responses of structural details and noise. By calibrating the frequency features with spatial structures, the model mitigates noise while preserving fine textures. Thus, by fusing the four flows, the TT block effectively bridges and reconciles the stochastic generative prior with deterministic HQS updates, ensuring high perceptual quality without sacrificing data fidelity. In addition, Diff-DUN employs stage-dependent denoising strengths to handle varying noise levels and adopts the more suitable v-prediction strategy for prior generation. The main contributions are highlighted below.

1) We propose Diff-DUN, an HQS-based unfolding framework for HSI denoising. It bridges and reconciles the observation consistency of DUNs with the perceptual quality of LDMs, and adaptively adjusts stagewise denoising strengths via learnable parameters.
2) We design a frequency-aware TT block as the core building block of the denoiser, which introduces explicit frequency flow and cross-flow interactions to jointly model spatial, spectral, and frequency characteristics for more faithful recovery.
3) Extensive experiments on synthetic and real HSI datasets confirm Diff-DUN's favorable restoration accuracy and computational efficiency (see Fig. 1). Ablation studies further quantify the contributions of each information flow and key model settings.

We organize the rest of this article as follows. Section II reviews related work on deep unfolding and generative priors. Section III presents the degradation-free prior generation and the TT block. Section IV reports experimental results on comprehensive datasets. Section V summarizes this article and outlines future directions.

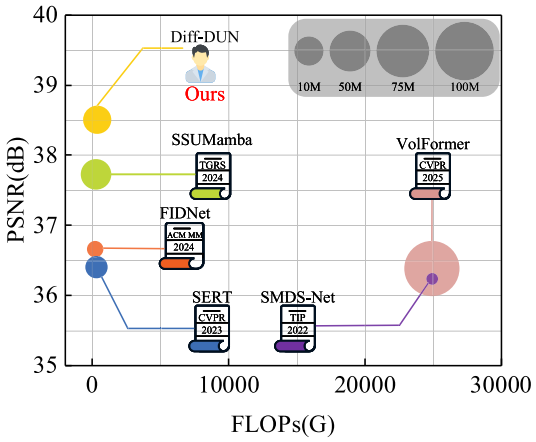


Fig. 1. Tradeoff among average PSNR, FLOPs, and parameter counts for recent supervised denoising methods under Case 1. PSNR is averaged over 562 test samples. Marker size represents the parameter count (M). Diff-DUN achieves the highest PSNR with moderate computational cost and model size.

II. Related Work

We review two research directions highly relevant to our work, namely DUNs and generative priors, both in the context of HSI denoising.

A. DUN for HSI Denoising

Historically, HSI denoising methods have evolved from model-based to DL methods. Model-based methods [6], [7], [8] offer interpretability but demand exhaustive manual parameter tuning. Conversely, DL methods [12], [13] achieve superior performance through powerful nonlinear representations. However, their black-box nature limits generalization to unseen noise distributions. DUNs [14], [15], [16], [17], [30], [31], [32] bridge these paradigms by unfolding convex optimization into learnable networks, integrating physical interpretability with data-driven flexibility. Early DUNs focus on unfolding sparse coding formulations. For instance, Xiong et al. [15] develop the subspace-based multidimensional sparse network (SMDS-Net) within the DUN to capture the spatial–spectral correlation, the low-rank property in the spectral dimension, and the spatial sparsity of HSIs. Along this line, Bodrito et al. [31] propose the trainable spatial–spectral sparse coding (T3SC) model, which unrolls a two-layer sparse coding process to explicitly characterize spectral unmixing and spatial–spectral correlations. Despite their promise, their reliance on handcrafted sparse priors limits their ability to handle complex noise patterns. To tackle this limitation, subsequent DUNs explore more flexible priors. Chen et al. [32] propose the deep low-rank decomposition model with flexible priors (Flex-DLD). This method employs the alternating direction method of multipliers (ADMMs) to integrate a self-supervised low-rank module into an optimization framework, effectively capturing learned and handcrafted priors. More recently, Transformer-based DUNs have further advanced performance. Zeng et al. [17] develop the degradation-noise-aware unfolding network (DNA-Net), the first DUN to employ Transformers for HSI denoising. By

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explicitly modeling noise distributions and capturing long-range spatial–spectral dependencies, DNA-Net demonstrates strong robustness in complex scenarios. To address the fixed-depth constraint, Ye et al. [16] develop the deep equilibrium convolutional sparse coding (DECSC) framework. It models hidden layers as a fixed-point solution to realize an effectively infinite-depth network with stronger convergence guarantees. Despite these advances, existing DUNs share a common limitation, i.e., their discriminative denoisers lead to the "regression to the mean" phenomenon [18]. As a result, they tend to produce oversmoothed outputs, especially under severe noise where high-frequency textures are progressively suppressed. On the contrary, stochastic generative models learn representative priors from data distributions and can generate boundaries that deterministic DUNs often suppress.

B. Generative Priors for HSI Denoising

Generative models, particularly diffusion-based methods [19], [29], [33], [34], offer an alternative by learning explicit data distributions rather than deterministic mappings. Early works integrate untrained or improved diffusion priors into the restoration pipeline. For instance, Miao et al. [21] propose the denoising diffusion spatio-spectral model (DDS2M), which employs untrained neural networks to characterize HSI priors within a diffusion model. Concurrently, Pang et al. [22] introduce an unsupervised HSI restoration method via improved diffusion models (HIR-Diff), which obtains HSIs as the product of reduced images and their associated coefficient matrices. Despite their effectiveness in utilizing structural properties, they still struggle to provide physically interpretable guidance. Therefore, another research direction integrates explicit physical modeling into LDMs. For instance, Zeng et al. [23] develop a self-supervised unmixing diffusion (Diff-Unmix) model for HSI denoising, which merges matrix factorization with a deep diffusion prior. Building on this idea, Wei et al. [35] introduce an unmixing-guided learnable abundance diffusion framework (ULADiff), which bridges physical modeling and generative processes. More recently, Li et al. [34] propose the latent diffusion enhanced rectangle Transformer (LDERT), which leverages an LDM to synthesize image-specific low-rank dictionaries, overcoming the representational limitations of traditional priors. Beyond the diffusion paradigm, alternative generative models have also been investigated. For example, Wang et al. [36] introduce the flow network for HSI denoising (HIDFlowNet), which characterizes denoising as a conditional distribution learning task. By leveraging an invertible architecture, it explicitly models the distribution of clean HSIs. However, these methods rely primarily on generative priors and are susceptible to hallucinations [21], [22], [23], [34], as diffusion models often generate content that deviates from the original input [24], [25], [26], [37]. Therefore, how to efficiently balance the stochastic nature of LDMs and strict data consistency remains a critical challenge for HSI denoising. Inspired by these complementary strengths [27], [28], [29], we propose to incorporate an LDM as a stochastic prior provider within a DUN, enabling the network to escape conservative estimates while maintaining observation consistency.

III. Latent Diff-DUN

A. Problem Formulation

Restoring a high-fidelity HSI from its noisy counterpart constitutes an inherently ill-posed inverse task. In this study, we represent the clean HSI as a third-order tensor $\mathcal{X} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C are the height, width, and number of spectral bands, respectively. The degradation process is typically modeled as

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} \tag{1}$$

with $\mathcal{Y}$ being the observed noisy HSI and $\mathcal{N}$ the additive Gaussian noise. Due to the ill-posed nature of the inverse problem in (1), recovering $\mathcal{X}$ from $\mathcal{Y}$ is infeasible without additional priors. Hence, we incorporate prior knowledge to constrain the solution space. To this end, we use a variational formulation that balances data fidelity with a prior term

$$\hat{\mathcal{X}} = \arg\min_{\mathcal{X}} \frac{1}{2}\left\|\mathcal{X} - \mathcal{Y}\right\|_2^2 + \tau\phi\left(\mathcal{X}\right) \tag{2}$$

where the data fidelity is enforced by the first term, and $\phi(\cdot)$ is a deep regularizer that encodes prior knowledge. The hyperparameter τ controls the tradeoff between the fidelity and regularization, and $\|\cdot\|_2$ denotes the ℓ_2-norm.

We leverage HQS [38] to handle this optimization. With an auxiliary variable $\mathcal{V}$ introduced, (2) is transformed into

$$(\mathcal{X}, \mathcal{V}) = \arg\min_{\mathcal{X},\mathcal{V}} \frac{1}{2}\left\|\mathcal{X} - \mathcal{Y}\right\|_2^2 + \frac{\rho}{2}\left\|\mathcal{X} - \mathcal{V}\right\|_2^2 + \tau\phi\left(\mathcal{V}\right) \tag{3}$$

where $\rho > 0$ is a penalty parameter. Equation (3) is then solved by alternately updating $\mathcal{X}$ and $\mathcal{V}$.

Update $\mathcal{X}$: At the kth HQS iteration, the $\mathcal{X}$-subproblem is given by

$$\mathcal{X}^{k+1} = \arg\min_{\mathcal{X}} \left\{\frac{1}{2}\left\|\mathcal{X} - \mathcal{Y}\right\|_2^2 + \frac{\rho}{2}\left\|\mathcal{X} - \mathcal{V}^k\right\|_2^2\right\} \tag{4}$$

which admits a closed-form solution

$$\mathcal{X}^{k+1} = \frac{\mathcal{Y} + \rho\mathcal{V}^k}{1+\rho}. \tag{5}$$

Update $\mathcal{V}$: Given $\mathcal{X}^{k+1}$, the $\mathcal{V}$-subproblem is expressed as

$$\mathcal{V}^{k+1} = \arg\min_{\mathcal{V}} \left\{\frac{\rho}{2}\left\|\mathcal{X}^{k+1} - \mathcal{V}\right\|_2^2 + \tau_k\phi\left(\mathcal{V}\right)\right\}. \tag{6}$$

Equation (6) is a Gaussian denoising problem with noise level $\sigma_k = (\tau_k/(\rho+\varepsilon))^{1/2}$, where ε is a small constant to prevent numerical instability. The corresponding proximal operator of the implicit prior $\phi(\cdot)$ is approximated by a learnable denoiser. Thus, the solution is given by the proximal operator

$$\mathcal{V}^{k+1} = \mathcal{D}_k\left(\mathcal{X}^{k+1}, Z, \sigma_k\right) \tag{7}$$

where $\mathcal{D}_k$ denotes the denoiser at the kth unfolding stage and Z denotes the target degradation-free prior Z_0 in Phase I and the generated prior $\hat{Z}_0$ in Phase II and inference. These definitions are detailed in Section III-C.

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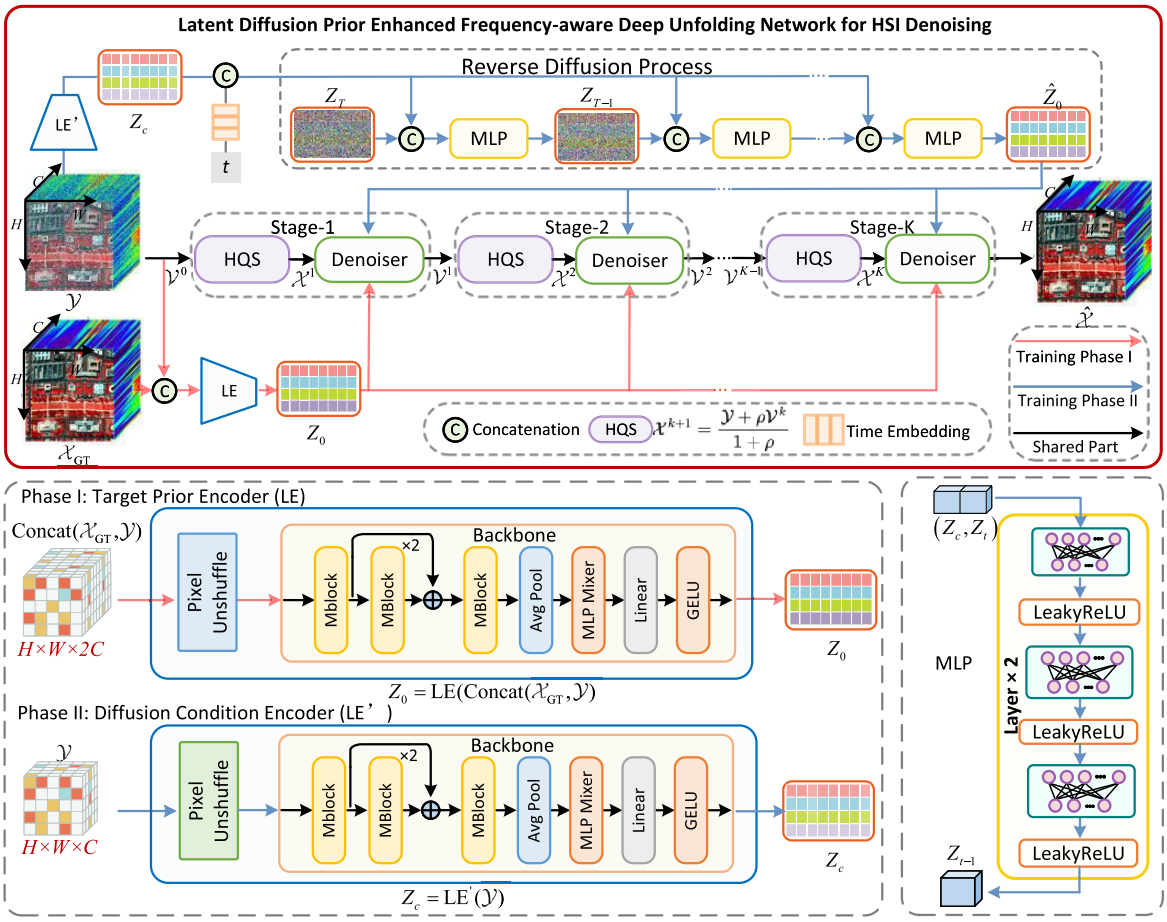


Fig. 2. Overall architecture of the proposed method. The framework employs a two-phase training strategy. In Phase I, the LE extracts a degradation-free prior Z_0 to serve as the training target. Phase II trains an LDM, conditioned on noisy observations, to generate this prior, which is then used by the DUN for iterative restoration. LE and LE′ are two different encoders with independent parameters. LE serves as the degradation-free prior target encoder, while LE′ serves as the conditional encoder.

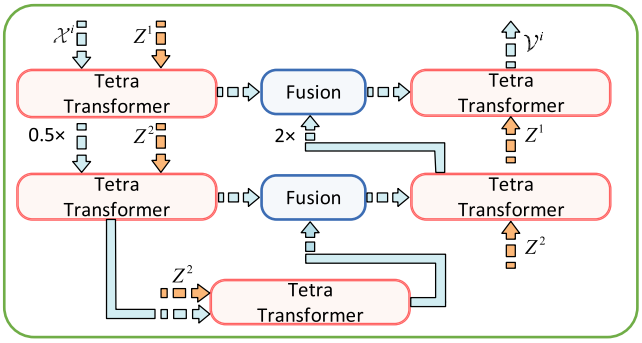


Fig. 3. Architecture of the denoiser. It adopts a U-shaped hierarchical structure [29] built around TT blocks, with the degradation-free prior Z^1 and its downsampling counterparts injected.

B. Network Architecture

As illustrated in Fig. 2, our framework unfolds the iterative optimization into a multistage architecture. At each stage, an HQS module ensures observation consistency, while a learnable proximal operator $\mathcal{D}_k$ driven by a degradation-resistant LDM performs high-quality denoising. In particular, as detailed in Fig. 3, this proximal mapping is realized via a U-shaped network [29] centered on a TT block. Unlike conventional spatial–spectral Transformers, our TT block leverages a degradation-free prior and its progressively downsampled multiscale representations to guide the denoising process.

In the following, we discuss two core modules of our framework, i.e., the generation of a degradation-free prior via an LDM and the prior-guided TT block within the denoiser.

C. Degradation-Free Prior Generation via an LDM

To counteract the performance bottleneck caused by the intrinsic ill-posedness in handling severe degradation, we introduce a degradation-free prior to guide the denoising process. However, extracting such a prior from noisy observations alone is challenging due to the entanglement of noise and clean signal. More critically, although ground truth (GT) is available during training, generating a degradation-free prior without GT guidance during the testing phase remains difficult. To address these issues [29], [34], [39], we employ a two-phase training strategy that conditions an LDM on degraded HSIs to produce the prior. In particular, in Phase I, we construct degradation-free prior samples by projecting GT into the latent space. In Phase II, we train an LDM to approximate this distribution, enabling prior generation from noisy observations

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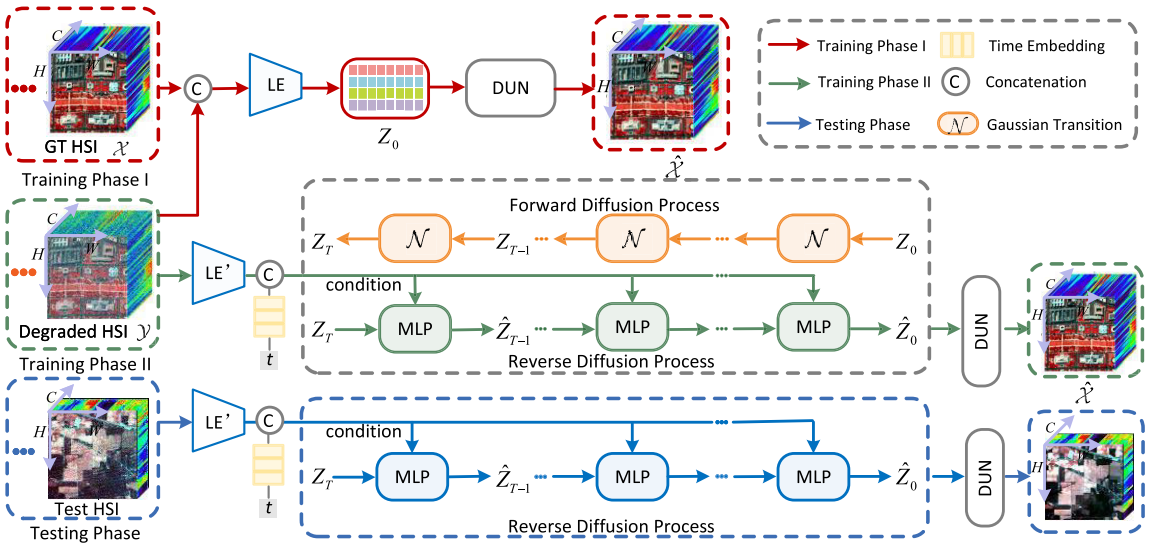


Fig. 4. Overview of the two-phase HSI denoising strategy. Phase I establishes a degradation-free prior from clean HSIs. Phase II trains an LDM, conditioned on noisy observations, to approximate this prior. During testing, the conditional LDM generates the prior from an unseen noisy HSI without access to GT, thereby facilitating iterative denoising.

without access to GT during the testing phase. Fig. 4 illustrates the overall pipeline.

Phase I (Learning Degradation-Free Prior From GT): Given the clean HSI $\mathcal{X}_{\text{GT}}$ and its noisy counterpart $\mathcal{Y}$, we obtain the target degradation-free prior Z_0 via

$$Z_0 = \text{LE}\left(\text{Concat}\left(\mathcal{X}_{\text{GT}}, \mathcal{Y}\right)\right) \tag{8}$$

where the latent encoder (LE) is a prior projection block. As shown in Fig. 2, LE [29] first applies pixel unshuffle to increase the channel dimension, followed by mobile blocks (MBlocks) [40] and MLP-Mixer [41] to capture local details and aggregate multidimensional information.

The training objective of LE in Phase I is to capture the prior representation Z_0, which serves as the degradation-free prior for the denoiser. Unlike conventional LDMs equipped with a "dedicated" decoder [42], Diff-DUN employs the DUN as the restoration decoder, in which Z_0 is injected to guide HSI denoising. In this phase, we jointly optimize LE and the DUN under the $\mathcal{L}_1$ loss

$$\mathcal{L}_1 = \left\|\hat{\mathcal{X}} - \mathcal{X}_{\text{GT}}\right\|_1 \tag{9}$$

where $\hat{\mathcal{X}}$ is the restored HSI. This joint training ensures that the learned prior captures information beneficial for denoising.

Phase II (Latent Diffusion Prior Generation and Joint Restoration): In Phase II, the parameters of LE are frozen, and LE is only used to generate the target degradation-free prior Z_0 for supervision. Meanwhile, an independent conditional encoder LE′ extracts the condition $Z_c = \text{LE}'(\mathcal{Y})$ from the noisy observation. Our objective is to generate a prior $\hat{Z}_0$ conditioned on Z_c, which then guides the DUN to reconstruct the restored HSI $\hat{\mathcal{X}}$. A straightforward supervision strategy for this prior generation is latent-space matching. We define this candidate objective and the image restoration loss as follows:

$$\mathcal{L}_{\text{reg}} = \left\|Z_0 - \hat{Z}_0\right\|_1 \tag{10}$$

$$\mathcal{L}_2^2 = \left\|\mathcal{X}_{\text{GT}} - \hat{\mathcal{X}}\right\|_1. \tag{11}$$

Rather than directly minimizing $\mathcal{L}_{\text{reg}}$ in (10), we learn prior generation through a conditional diffusion process with velocity prediction while retaining $\mathcal{L}_2^2$ for image restoration. In particular, the forward diffusion process progressively adds Gaussian noise to Z_0 over T steps with a fixed variance schedule $\{\beta_t\}_{t=1}^T$, producing the Markov chain $Z_0, Z_1, \ldots, Z_T$

$$q\left(Z_t \mid Z_{t-1}\right) = \mathcal{N}\left(Z_t;\ \sqrt{1-\beta_t}Z_{t-1}, \beta_t\mathbf{I}\right) \tag{12}$$

with Z_t denoting the noisy latent variable at step t, and Z_0 the generative target. By the standard reparameterization trick, the distribution $q(Z_t \mid Z_0)$ admits a closed-form expression

$$q\left(Z_t \mid Z_0\right) = \mathcal{N}\left(Z_t;\ \sqrt{\bar{\alpha}_t}Z_0, (1-\bar{\alpha}_t)\,\mathbf{I}\right) \tag{13}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$.

With the forward diffusion defined, we next consider the reverse process. Here, our goal is to recover Z_0 from a standard Gaussian latent variable Z_T, conditioned on Z_c. Unlike conventional ϵ-prediction, which directly estimates the added noise, the v-prediction [43], [44] strategy defines the velocity as a time-dependent linear combination of the clean latent prior Z_0 and the Gaussian noise ϵ_t. In particular, given the forward noise $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$, the velocity v_t at step t is defined as

$$v_t = \sqrt{\bar{\alpha}_t}\,\epsilon_t - \sqrt{1-\bar{\alpha}_t}\,Z_0. \tag{14}$$

This parameterization allows direct recovery of both Z_0 and ϵ_t from the predicted velocity $\hat{v}_t = v_\theta(Z_t, Z_c, t)$

$$\hat{Z}_0 = \sqrt{\bar{\alpha}_t}\,Z_t - \sqrt{1-\bar{\alpha}_t}\,\hat{v}_t \tag{15}$$

$$\hat{\epsilon}_t = \sqrt{1-\bar{\alpha}_t}\,Z_t + \sqrt{\bar{\alpha}_t}\,\hat{v}_t \tag{16}$$

leading to the reverse update rule

$$Z_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left[Z_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\left(\sqrt{1-\bar{\alpha}_t}Z_t + \sqrt{\bar{\alpha}_t}\hat{v}_t\right)\right]. \tag{17}$$

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The conditional diffusion model is trained by optimizing the following velocity prediction loss:

$$\mathcal{L}_2^1 = \mathbb{E}_{t,Z_0,\epsilon_t}\left[\left\| v_t - v_\theta\left(Z_t, Z_c, t\right)\right\|_2^2\right]. \tag{18}$$

Given the low-dimensional nature of the prior (16×256), we replace the heavy U-Net backbone with a lightweight MLP for v_θ. The LDM performs the full reverse process with $T = 16$ steps to synthesize $\hat{Z}_0$. Throughout Phase II, LE remains frozen, while LE′, the DUN, and v_θ are jointly optimized. The total training objective is

$$\mathcal{L}_2 = \mathcal{L}_2^1 + \mathcal{L}_2^2 \tag{19}$$

where $\mathcal{L}_2^2 = \|\mathcal{X}_{\mathrm{GT}} - \hat{\mathcal{X}}\|_1$ ensures joint restoration quality. Algorithm 1 summarizes the complete two-phase procedure.

Algorithm 1 Two-Phase Training Strategy

Require: Dataset $\mathcal{D} = \{(\mathcal{X}_{\mathrm{GT}}^{(m)}, \mathcal{Y}^{(m)})\}_{m=1}^{M}$; randomly initialized parameters of DUN ϕ_{DUN}, latent encoder LE network ϕ_{LE}, conditional encoder LE′ network ϕ_{CLE}, and diffusion network LDM ϕ_v.

while not converged **do** ▷Phase I training
  $Z_0 \leftarrow \mathrm{LE}(\mathrm{Concat}(\mathcal{X}_{\mathrm{GT}}, \mathcal{Y}); \phi_{\mathrm{LE}})$
  $\hat{\mathcal{X}} \leftarrow \mathrm{DUN}(\mathcal{Y}, Z_0; \phi_{\mathrm{DUN}})$
  Update ϕ_{DUN} and ϕ_{LE} using $\mathcal{L}_1$
end while
Freeze ϕ_{LE}
while not converged **do** ▷Phase II training
  $Z_0 \leftarrow \mathrm{LE}(\mathrm{Concat}(\mathcal{X}_{\mathrm{GT}}, \mathcal{Y}); \phi_{\mathrm{LE}})$ ▷Generated by the frozen LE
  $Z_c \leftarrow \mathrm{LE}'(\mathcal{Y}; \phi_{\mathrm{CLE}})$
  $\hat{Z}_0 \leftarrow \mathrm{LDM}(Z_c; \phi_v)$
  $\hat{\mathcal{X}} \leftarrow \mathrm{DUN}(\mathcal{Y}, \hat{Z}_0; \phi_{\mathrm{DUN}})$
  Update ϕ_{DUN}, ϕ_{CLE}, and ϕ_v using $\mathcal{L}_2$
end while

D. Prior-Guided TT Within the Denoiser

While the LDM provides degradation-free priors with rich details, effectively incorporating them into the DUN's denoising process remains nontrivial. Previous denoisers typically rely solely on spatial–spectral correlations, which are inherently difficult to capture from noisy observations alone. Moreover, spatial and spectral flows operate in the pixel domain, failing to capture the underlying structural patterns latent in raw pixel values. To address these issues, we introduce a frequency flow that unveils such fundamental patterns via Fourier-domain analysis [45]. Consequently, we design the TT block that aggregates four types of complementary information, i.e., the degradation-free prior flow, the spectral, spatial, and frequency flows. Among them, the spatial flow leverages local pixel correlation to remove noise across different regions. Concurrently, the spectral flow models spectral correlations to counteract spectral perturbations caused by the noise. Subsequently, the observation features from both the spatial and spectral flows serve as guidance to select observation-consistent information from the degradation-free prior adaptively. This strategy helps reduce artifacts introduced by the prior, thereby ensuring fidelity to the observed HSIs. In addition, we introduce an explicit frequency flow to capture the differential frequency responses between structural details and noise components. By calibrating the frequency features with spatial structures, the model mitigates noise while preserving fine textures and sharp edges. Consequently, by fusing the four flows, the TT block effectively bridges and reconciles the stochastic generative prior with deterministic HQS updates, jointly ensuring high perceptual quality and data fidelity.

In particular, as shown in Fig. 5, we illustrate the information flow using the initial TT block within the U-shaped denoiser as an example. The input $\mathcal{X}^i$ is simultaneously routed to the spatial, spectral, and frequency flows. In addition, the prior flow receives the degradation-free prior Z_0 or $\hat{Z}_0$ described in Section III-C to guide the denoising process.

1) Prior Flow: To effectively incorporate the degradation-free prior, we employ a multiscale hierarchical prior representation inspired by [29], [46], and [47]. As formulated in (20), we progressively downsample the initial degradation-free prior (either Z_0 in Phase I or $\hat{Z}_0$ in Phase II) to match the U-shaped denoiser in Fig. 3. The multiscale priors are defined as

$$Z^i = \begin{cases} Z_0 \text{ or } \hat{Z}_0, & if\ i = 1 \\ \mathrm{downsample}\left(Z^{i-1}\right), & if\ i > 1 \end{cases} \tag{20}$$

where $Z^i \in \mathbb{R}^{N/(4^{i-1}) \times C_z}$, $i = 1, 2$, $N = 16$, and $C_z = 256$. For Phase I, $Z^1 = Z_0$, whereas for Phase II and inference, $Z^1 = \hat{Z}_0$.

Beyond the multiscale design, as illustrated in Fig. 5(a), the prior flow is built upon a multihead cross-attention (MHCA) mechanism. In particular, this flow employs spatially aware features from the spectral flow as the Q, while K and V are from the degradation-free prior. Through the above design, we enable the degradation-free prior to be selectively incorporated based on spatial–spectral characteristics and compensate for missing spatial and spectral information.

2) Spectral Flow: As demonstrated in Fig. 5(b), the spectral flow employs a multihead self-attention (MHSA) mechanism to capture spectral correlations. However, due to the high redundancy of HSIs, directly applying MHSA to HSIs is computationally expensive. To mitigate this issue [13], [29], we reduce the resolution of the embeddings by convolutions and simultaneously double their channel dimensions [48]. To compensate for the inevitable loss caused by downsampling, we introduce a spatial–spectral modulation module, which performs elementwise multiplication between the downsampled embeddings and local spatial features borrowed from the spatial flow. As a result, the spectral flow captures the spectral correlation with spatial awareness.

3) Spatial Flow: The spatial flow, as shown in Fig. 5(c), employs lightweight MBlocks to capture local pixel correlation. Each MBlock adopts an inverted residual structure with depthwise convolutions, wherein batch normalization is intentionally excluded to maintain low computational complexity. In addition, to explore the relationship between channels, we further employ squeeze and excitation (SE) [49] to recalibrate channelwise feature responses. Consequently, the spatial flow effectively captures local spatial correlations at a markedly reduced computational cost.

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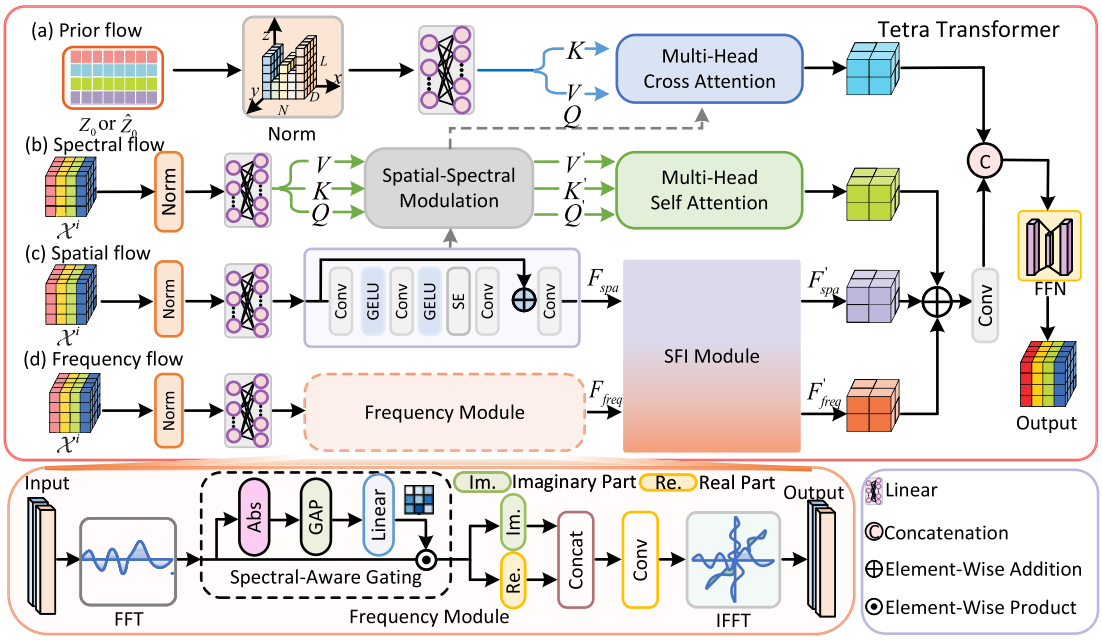


Fig. 5. Detailed architecture of the TT block. The TT block aggregates four types of complementary information. (a) Degradation-free prior flow. (b) Spectral flow. (c) Spatial flow. (d) Frequency flow. This multiflow synergy ensures robust feature representation for HSI denoising.

4) Frequency Flow: The frequency flow is motivated by the observation that HSI noise exhibits distinct energy distributions across the Fourier spectrum. To this end, we decouple input features into the frequency domain. As depicted in Fig. 5(d), for an input feature map x, the 2-D FFT produces $\mathcal{F}(x) = R(x) + jI(x)$ with amplitude $\mathcal{A}(x) = (R^2(x) + I^2(x))^{1/2}$, where $R(\cdot)$ and $I(\cdot)$ are the real and imaginary parts. To adaptively emphasize informative frequency components, we introduce a spectrum-aware gating (SAG) mechanism

$$\hat{\mathcal{F}}(x) = \mathcal{F}(x) \odot \text{Sigmoid}(\text{Linear}(\text{GAP}(\mathcal{A}(x)))) \tag{21}$$

where GAP denotes global average pooling. The real and imaginary parts of $\hat{\mathcal{F}}(x)$ are concatenated into F_{ri}, then processed by 3×3 depthwise and 1×1 pointwise convolutions. The result is projected back to the spatial domain via inverse FFT. To enable bidirectional interaction between local and global features, the spatial-frequency interaction (SFI) module exchanges information between the spatial and frequency flows through intraflow and cross-flow convolutions.

5) Cross-Flow Interaction and Aggregation: To fully exploit the correlations among the four flows, the TT block, in addition to the aforementioned interaction, enables interflow interaction via lightweight 1×1 residual convolutional bridges. These features are then fused and injected into a feed-forward network (FFN) [13], yielding the refined output.

IV. Experiments

To comprehensively evaluate our method, extensive experiments are conducted on simulated and real-world HSI datasets. We detail the experimental settings, compare against state-of-the-art (SOTA) approaches, and conduct ablation studies to quantify the contribution of each core module.

A. Experimental Settings

1) Datasets: We assess the proposed method on both simulated and real-world HSI datasets. In particular, we employ four benchmark datasets, namely the CAVE,[1] KAIST,[2] ICVL,[3] and Harvard University[4] datasets for simulated experiments.

1) CAVE [55] contains 32 HSIs, each with 31 spectral bands covering 400–700 nm and a spatial resolution of 512×512. We use 25 images for training and the remaining seven for testing.
2) KAIST [56] is a high-resolution HSI dataset comprising 30 images with size 2704×3376, each featuring 31 spectral bands ranging from 420 to 720 nm. We adopt 20 images for training and ten for testing.
3) ICVL [57] contains 201 HSIs with a spatial resolution of 1392×1300 and 31 spectral channels. We randomly partition the dataset into 100 training and 50 testing images.
4) The Harvard dataset contains 50 HSIs, each of size 1392×1040 with 31 spectral channels spanning 420–720 nm. We employ 30 images for training and ten for testing.

For preprocessing, all images are first normalized to [0, 1] and then cropped into nonoverlapping patches of size 256×256 with 28 spectral bands. This yields 4138 training patches and 562 testing patches, with strict separation between training and testing sets to ensure fair evaluation.

2) Evaluation Metrics: Three standard metrics, i.e., peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and erreur relative globale adimensionnelle de synthése (ERGAS),

[1] http://www1.cs.columbia.edu/CAVE/databases/multispectral/
[2] https://vclab.kaist.ac.kr/siggraphasia2017p1/kaistdataset.html
[3] https://icvl.cs.bgu.ac.il/hyperspectral/
[4] http://vision.seas.harvard.edu/hypers

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TABLE I

QUANTITATIVE COMPARISON ON REPRESENTATIVE SAMPLES FROM FOUR HSI DATASETS UNDER TWO I.I.D. GAUSSIAN NOISE CASES. RESULTS ARE OBTAINED USING THE THREE-STAGE DIFF-DUN. THE BEST AND SECOND-BEST VALUES ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Dataset	Case	Metric	Observed	BM4D [5]	NGMeet [50]	SMDS-Net [15]	HIR-Diff [22]	FIDNet [51]	DRO-TFF [7]	VolFormer [52]	Diff-DUN
4cam_0411-1640-1_patch_01 (ICVL dataset[3])	Case 1	PSNR ↑	10.4615	33.0464	37.4812	36.0283	35.8242	38.3780	37.2688	37.9924	**40.0985**
		SSIM ↑	0.0196	0.8693	0.9614	0.9172	0.9364	0.9659	0.9567	0.9453	**0.9718**
		ERGAS ↓	1130.4	95.6710	43.1793	50.8961	53.9652	68.6710	44.3807	40.7044	**31.6406**
	Case 2	PSNR ↑	13.9834	35.3623	40.2447	39.2351	37.7556	40.9476	38.6369	39.4706	**41.6760**
		SSIM ↑	0.0420	0.9156	0.9719	0.9674	0.9289	0.9738	0.9640	0.9553	**0.9776**
		ERGAS ↓	753.6264	70.1638	31.5911	35.4817	41.7397	28.7411	38.2469	34.5028	**26.3463**
chart_and_stuffed_toy_patch_01 (CAVE dataset[1])	Case 1	PSNR ↑	10.4615	30.9013	**34.5978**	31.6027	33.4682	30.1498	32.2846	33.4080	34.3255
		SSIM ↑	0.0658	0.8736	**0.9621**	0.8847	0.9331	0.9220	0.9100	0.9332	0.9600
		ERGAS ↓	949.0561	90.4562	**59.1836**	83.2289	67.4665	98.8347	77.1530	67.8355	62.7624
	Case 2	PSNR ↑	13.9834	33.3158	**36.1715**	32.8280	36.0151	31.3448	34.2430	35.4929	36.1665
		SSIM ↑	0.1144	0.9300	**0.9750**	0.9425	0.9584	0.9377	0.9378	0.9539	0.9727
		ERGAS ↓	632.7040	68.6884	**43.9398**	72.2858	50.4364	89.5025	61.6425	53.5044	49.7832
KAIST_scene25_patch_004 (KAIST dataset[2])	Case 1	PSNR ↑	10.4615	31.2065	33.2113	30.9955	31.0565	32.7173	31.6456	32.5873	**33.3338**
		SSIM ↑	0.0556	0.8656	0.9204	0.8607	0.8807	0.9167	0.8920	0.9046	**0.9226**
		ERGAS ↓	920.4402	85.5211	67.3056	86.7055	87.3353	71.4995	81.1727	72.5772	**66.3930**
	Case 2	PSNR ↑	13.9834	32.8490	34.0862	32.3045	32.2178	33.9305	32.9406	33.6584	**34.1104**
		SSIM ↑	0.1076	0.8994	0.9274	0.9050	0.9014	0.9255	0.9085	0.9145	**0.9288**
		ERGAS ↓	613.6268	70.8576	**59.9618**	74.8963	75.9233	61.8858	69.5629	63.8930	60.7229
imga1_patch_03 (Harvard University dataset[4])	Case 1	PSNR ↑	10.4615	30.6172	35.7721	34.8282	34.4133	35.4105	34.7900	35.4066	**37.0094**
		SSIM ↑	0.0319	0.7838	0.9074	0.8873	0.8818	0.9104	0.8922	0.8980	**0.9253**
		ERGAS ↓	1108.9	100.7456	50.7211	55.8322	59.6017	53.0723	56.7690	52.4587	**43.2838**
	Case 2	PSNR ↑	13.9834	32.7647	37.8764	37.1815	36.3214	37.2091	36.0426	37.0203	**38.4811**
		SSIM ↑	0.0651	0.8444	0.9319	0.9270	0.9133	0.9283	0.9130	0.9179	**0.9382**
		ERGAS ↓	739.2766	76.6961	39.4869	42.4954	47.1023	42.7472	49.0132	43.3976	**36.7517**

are employed to quantitatively evaluate all methods. PSNR and SSIM measure spatial fidelity, while ERGAS quantifies spectral distortion. Accordingly, higher PSNR and SSIM values and lower ERGAS indicate better denoising performance, with ideal values of $+\infty$, 1, and 0, respectively.

3) Implementation Details: We optimize Diff-DUN using Adam with a batch size of 2. The training process is conducted in two phases. Phase I is performed for 100 epochs with an initial learning rate of 2×10^{-4} to learn the degradation-free prior. In Phase II, the model is jointly optimized for diffusion prior generation and HSI restoration for 100 epochs with an initial learning rate of 1×10^{-4}. In both phases, the learning rate is adjusted using a linear warm-up followed by cosine decay. The LDM employs a linear noise schedule with $T = 16$ diffusion steps and v-prediction strategy. The degradation-free prior has a size of 16×256, corresponding to $N = 16$ latent tokens with a feature dimension of $C_z = 256$.

4) Competing Methods: Our method is compared against SOTA methods, including BM4D [5], NGMeet [50], SMDS-Net [15], HIR-Diff [22], FIDNet [51], SERT [53], DDS2M [21], SSUMamba [54], DRO-TFF [7], and VolFormer [52]. Regarding the computational platforms, the model-based methods are implemented in MATLAB and run on an Intel i7-13700KF CPU, whereas all DL-based methods are evaluated on an NVIDIA GeForce RTX 4090 GPU. To ensure a fair comparison, we categorize the competing methods into three groups. The supervised DL methods, including SMDS-Net, FIDNet, SERT, SSUMamba, and VolFormer, are trained and evaluated under the same settings as Diff-DUN. Among the unsupervised methods, HIR-Diff operates as a sampling-based pretrained diffusion model, while DDS2M and DRO-TFF follow their image-specific self-supervised optimization strategies. BM4D and NGMeet employ the parameter settings recommended in their original papers. In terms of efficiency, HIR-Diff requires only 20 reverse sampling steps, whereas DDS2M needs approximately 2000 steps. The synthetic Gaussian noise is considered under four different cases, which we detail next.

5) Noise Cases: Case 1 (i.i.d. Gaussian, $\sigma = 0.3$): All spectral bands of the normalized HSI are degraded by i.i.d. Gaussian noise with a standard deviation of $\sigma = 0.3$.

Case 2 (i.i.d. Gaussian, $\sigma = 0.2$): Similarly, i.i.d. Gaussian noise with a standard deviation of $\sigma = 0.2$ is imposed on each band to simulate a lower noise level.

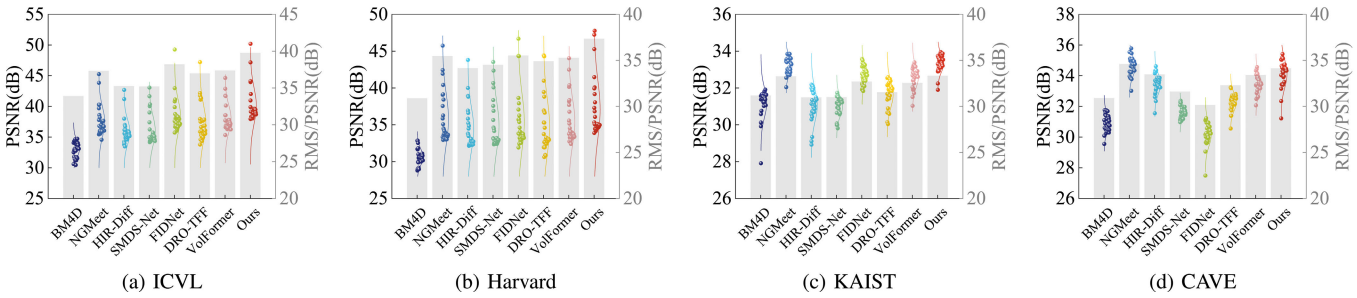
Case 3 (Bandwise Non-i.i.d. Gaussian Noise): Each spectral band is corrupted by zero-mean Gaussian noise with a band-dependent standard deviation. In particular, for the cth band, the noise is modeled as $\mathbf{N}_c \sim \mathcal{N}(0, \sigma_c^2)$, where σ_c is independently sampled from {10, 20, 30, 40, 50}/255.

Case 4 (Severe Bandwise Non-i.i.d. Gaussian Noise): Case 4 follows the setting of Case 3, but σ_c is sampled from {50, 60, 70, 80, 90}/255 randomly.

B. Synthetic Noise Experiment

1) Quantitative Comparison: Table I illustrates the comparison of the quantitative performance of different methods on representative samples from four HSI datasets under Cases 1 and 2. We highlight the best and second-best results for each metric in bold and underlined, respectively. As shown in Table I, Diff-DUN consistently ranks among the top two across the three evaluation metrics and both noise levels. Notably, it achieves clear improvements on the ICVL and Harvard University datasets while remaining highly competitive on the CAVE and KAIST datasets. This dataset-dependent variation may be associated with differences in the training-sample distributions [58]. Overall, these representative results

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(a) ICVL (b) Harvard (c) KAIST (d) CAVE

Fig. 6. Bandwise PSNR values of the denoising results under Case 1. (a) *4cam_0411-1640-1_patch_01* from the ICVL dataset. (b) *imga1_patch_03* from the Harvard dataset. (c) *KAIST_scene25_patch_004* from the KAIST dataset. (d) *chart_and_stuffed_toy_patch_01* from the CAVE dataset.

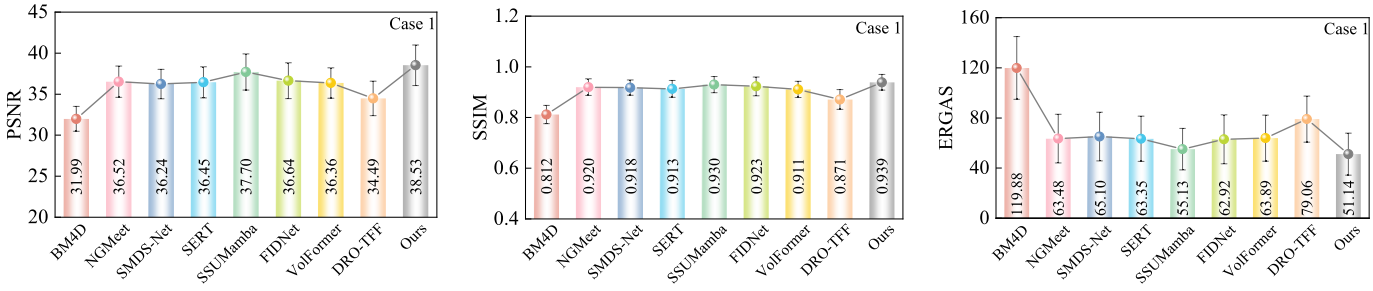


Fig. 7. Comparison of the average PSNR, SSIM, and ERGAS values of different methods over 562 test samples under Case 1.

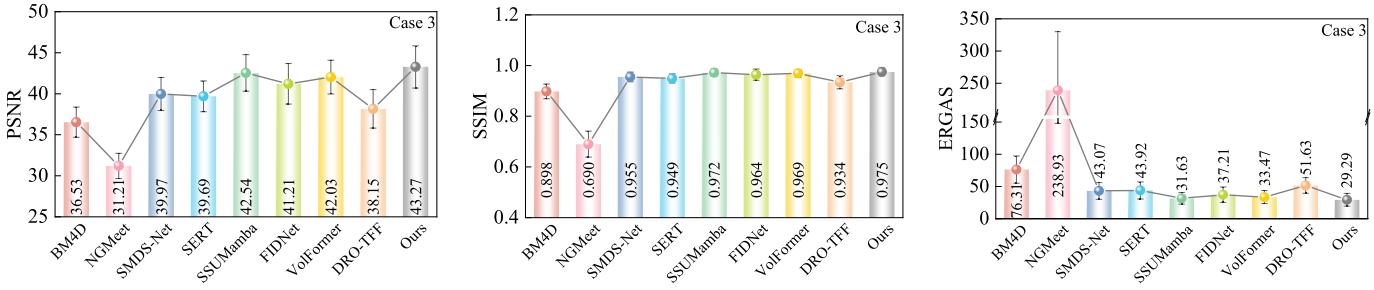


Fig. 8. Comparison of the average PSNR, SSIM, and ERGAS values of different methods over 562 test samples under Case 3.

demonstrate the robustness of Diff-DUN across different datasets and degradation levels. On the contrary, existing methods such as BM4D, SMDS-Net, and HIR-Diff perform well under low noise levels, but they tend to struggle in high-noise scenarios. To examine whether this performance is consistently maintained across spectral bands, Fig. 6 presents the bandwise PSNR results under Case 1. The dots denote the PSNR values of individual bands, while the bars represent their root mean square (RMS), computed as $\text{RMS} = \left(1/C \sum_{i=1}^{C} \text{PSNR}_i^2\right)^{1/2}$, where C is the number of spectral bands. As shown in Fig. 6, Diff-DUN achieves consistently high PSNR values across spectral bands for most representative samples under Case 1. Since the results for individual HSIs may not fully reflect the overall performance, Table II further reports the mean and standard deviation over 562 test samples and focuses on comparisons with recent supervised deep learning methods SMDS-Net, FIDNet, SERT, SSUMamba, and VolFormer. Therefore, Table II provides a complementary statistical evaluation under four noise cases. As summarized in Table II and visualized in Figs. 7 and 8, Diff-DUN consistently achieves the best restoration performance under both the i.i.d. and the more realistic band-dependent non-i.i.d. Gaussian noise settings. We also note that NGMeet performs poorly in Case 3, likely due to its assumption of a single global noise level across all bands, which does not hold under our band-dependent noise setting.

2) Qualitative Comparison: To qualitatively evaluate the denoising performance, we provide pseudo-RGB visualizations of all competing methods in Figs. 9 and 10. Note that below each image, we display a magnified view of a selected patch and the corresponding error map (i.e., the absolute difference from the GT). From these results, we observe that all methods effectively remove severe Gaussian noise. However, a meticulous inspection reveals significant differences in detail preservation. For example, BM4D tends to oversmooth sharp edges and fine textures, resulting in blurred structures in the magnified region. SMDS-Net and HIR-Diff while retaining more textures, occasionally introduce subtle artifacts or spectral distortions, which are reflected in the corresponding absolute error maps. On the contrary, our method, in addition to superior quantitative performance, achieves impressive visual fidelity by preserving the spatial textures

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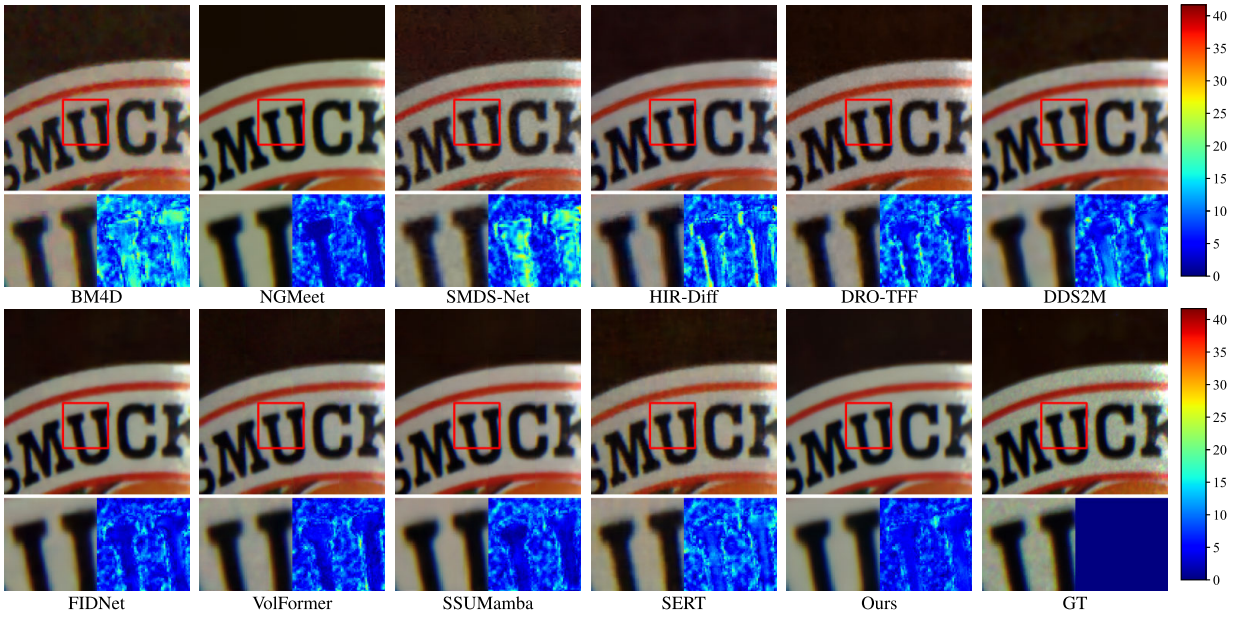


Fig. 9. Visualization of false-color denoising results on *KAIST_scene25_patch_004* (R:27, G:15, and B:4) under Case 1. For each result, we display the magnified region of a selected patch and its corresponding absolute error map relative to the GT.

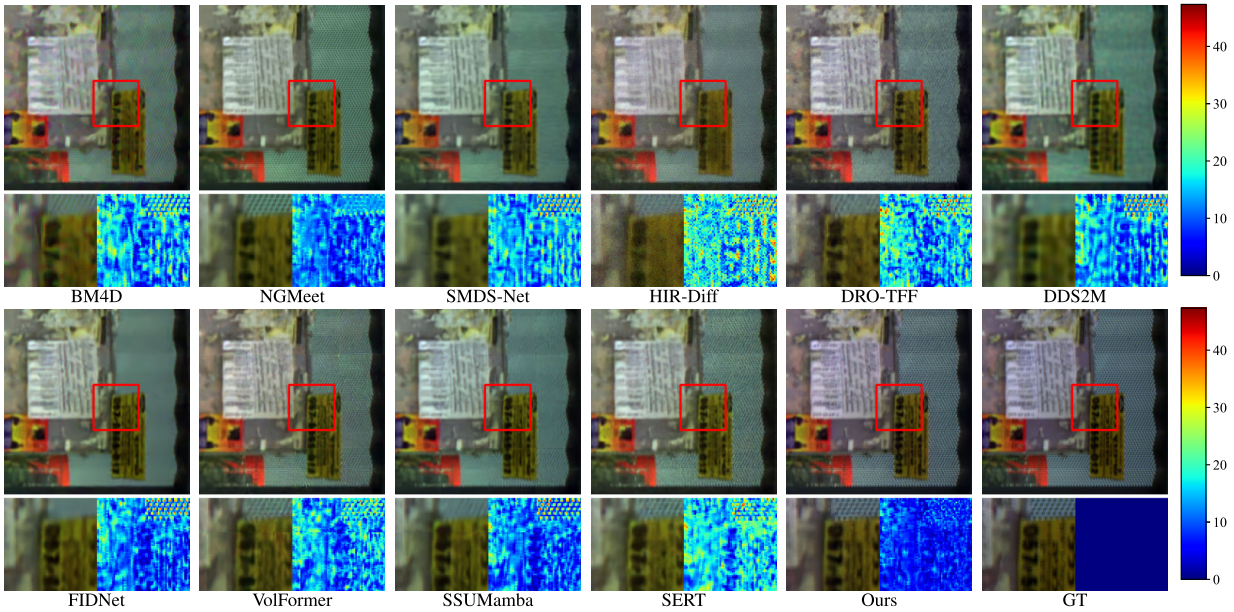


Fig. 10. Visualization of false-color denoising results on *rsh_0406-1443_patch_27* (R:26, G:18, and B:9) under Case 2. Each result includes a magnified patch and the corresponding absolute error map relative to the GT.

TABLE II

AVERAGE QUANTITATIVE COMPARISON OF DIFFERENT METHODS UNDER FOUR NOISE CASES ON 562 TEST SAMPLES. THE RESULTS ARE REPORTED IN THE FORM OF MEAN ± STANDARD DEVIATION. THE BEST AND SECOND-BEST VALUES ARE INDICATED IN BOLD AND UNDERLINED, RESPECTIVELY

Case	Metric	BM4D [5]	NGMeet [50]	SMDS-Net [15]	SERT [53]	SSUMamba [54]	FIDNet [51]	VolFormer [52]	DRO-TFF [7]	Diff-DUN
Case 1	PSNR ↑	31.99 ± 1.52	36.52 ± 1.90	36.24 ± 1.79	36.45 ± 1.90	37.70 ± 2.21	36.64 ± 2.18	36.36 ± 1.85	34.49 ± 2.10	**38.53 ± 2.47**
	SSIM ↑	0.812 ± 0.036	0.920 ± 0.033	0.918 ± 0.030	0.913 ± 0.033	0.930 ± 0.032	0.923 ± 0.037	0.911 ± 0.032	0.871 ± 0.039	**0.939 ± 0.031**
	ERGAS ↓	119.88 ± 24.99	63.48 ± 19.40	65.10 ± 19.39	63.35 ± 18.01	55.13 ± 16.54	62.92 ± 19.53	63.89 ± 18.37	79.06 ± 18.35	**51.14 ± 16.74**
Case 2	PSNR ↑	34.18 ± 1.64	38.69 ± 1.97	37.92 ± 1.82	38.63 ± 2.00	39.31 ± 2.23	38.32 ± 2.29	38.06 ± 1.85	36.35 ± 2.15	**40.21 ± 2.49**
	SSIM ↑	0.870 ± 0.030	0.944 ± 0.025	0.937 ± 0.023	0.942 ± 0.023	0.947 ± 0.025	0.941 ± 0.029	0.933 ± 0.024	0.908 ± 0.031	**0.955 ± 0.023**
	ERGAS↓	90.61 ± 19.06	49.29 ± 14.96	53.71 ± 15.91	49.18 ± 13.86	45.70 ± 13.66	51.58 ± 15.98	52.42 ± 14.80	63.70 ± 15.93	**41.69 ± 13.50**
Case 3	PSNR ↑	36.53 ± 1.85	31.21 ± 1.54	39.97 ± 2.01	39.69 ± 1.88	42.54 ± 2.22	41.21 ± 2.48	42.03 ± 2.05	38.15 ± 2.34	**43.27 ± 2.58**
	SSIM ↑	0.898 ± 0.029	0.690 ± 0.051	0.955 ± 0.019	0.949 ± 0.018	0.972 ± 0.014	0.964 ± 0.022	0.969 ± 0.015	0.934 ± 0.026	**0.975 ± 0.015**
	ERGAS↓	76.31 ± 21.18	238.93 ± 90.93	43.07 ± 13.11	43.92 ± 13.26	31.63 ± 9.04	37.21 ± 12.01	33.47 ± 10.11	51.63 ± 12.07	**29.29± 9.82**
Case 4	PSNR ↑	32.46 ± 1.59	30.72± 1.62	36.46 ± 1.81	36.68 ± 1.91	38.17 ± 2.24	37.17± 2.23	37.09 ± 1.94	34.83 ± 2.08	**39.00 ± 2.52**
	SSIM ↑	0.824 ± 0.038	0.719 ± 0.053	0.920 ± 0.030	0.916 ± 0.031	0.935 ± 0.030	0.929 ± 0.034	0.922 ± 0.031	0.879 ± 0.037	**0.944 ± 0.030**
	ERGAS↓	115.04 ± 24.95	225.37 ± 91.64	63.87 ± 19.17	61.78 ± 17.78	52.29 ± 15.65	59.06 ± 18.32	58.93 ± 17.12	76.09 ± 18.08	**48.47 ± 16.02**

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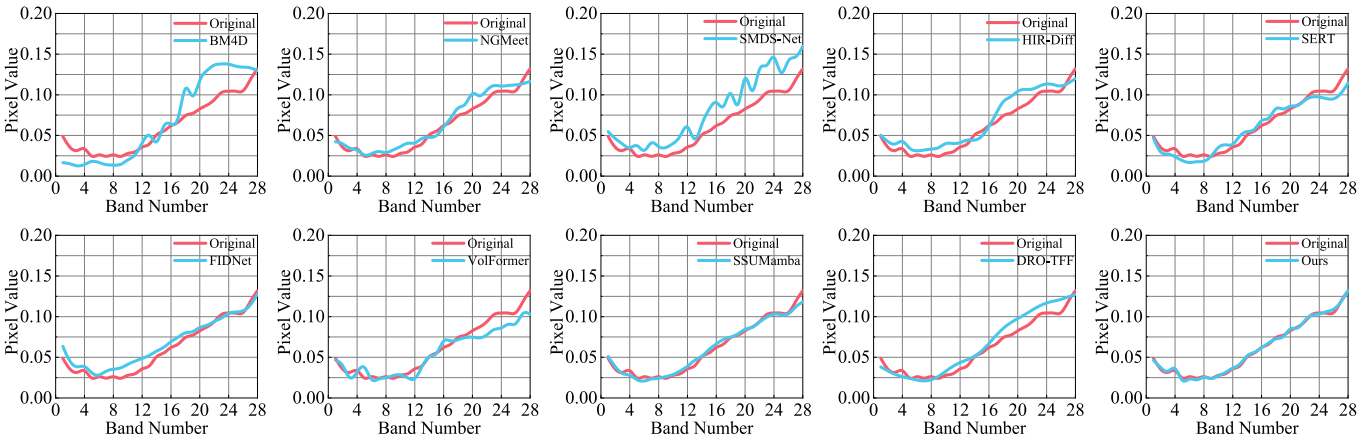


Fig. 11. Example spectral curves at the selected spatial location ($H = 57, W = 228$) of *KAIST_scene25_patch_004* under Case 1.

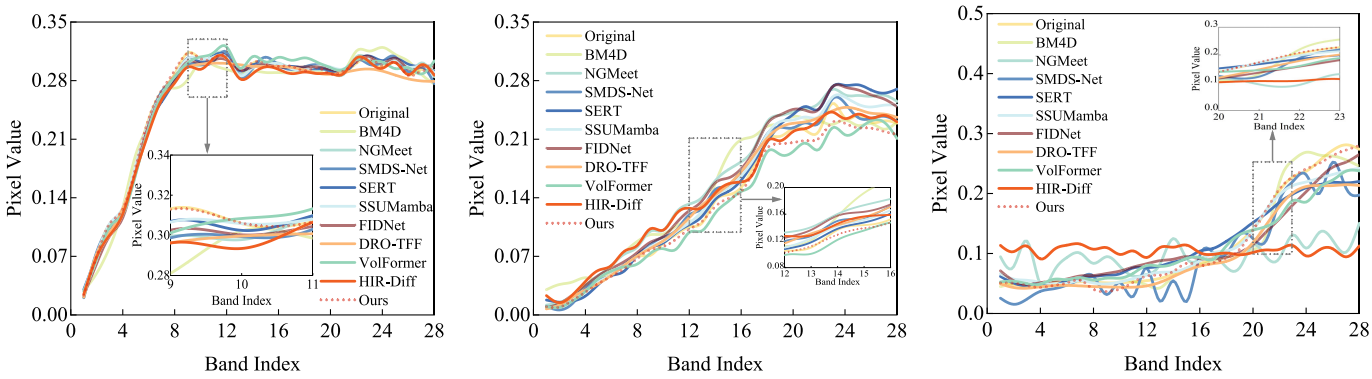


Fig. 12. Example spectral curves at selected spatial locations ($H = 38, W = 155$) of *4cam_0411-1640-1_patch_01*, ($H = 172, W = 97$) of *imga1_patch_03*, and ($H = 223, W = 253$) of *chart_and_stuffed_toy_patch_01* under Case 1.

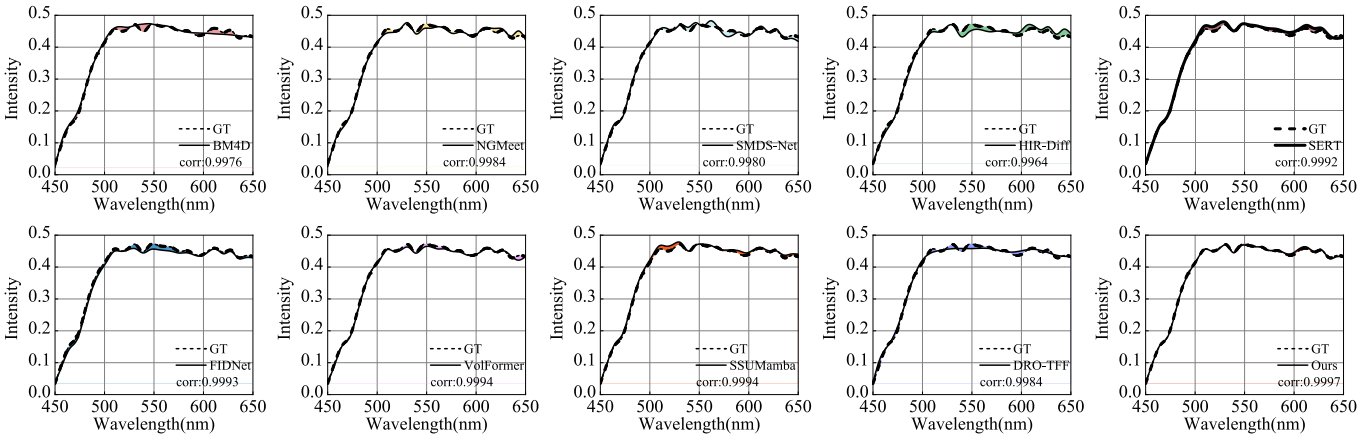


Fig. 13. Spectral intensity curves of the GT and restored results from different methods at a selected region of *4cam_0411-1640-1_patch_01* under Case 1. The value labeled "Corr" in the bottom-right corner indicates the correlation coefficient between each restored curve and the GT curve.

and structural details of the original HSIs. Beyond the pseudo-RGB visualizations, we further evaluate the spectral fidelity by presenting the spectral curve recovery at the selected locations for all competing methods under Case 1 in Figs. 11 and 12. Meanwhile, we calculate the correlation coefficient ("Corr") between the restored and GT spectra extracted from the same selected region. As evidenced by the "Corr" values in Fig. 13, Diff-DUN achieves spectral intensity curves that align more closely with the GT than those recovered by the other methods.

3) Frequency-Domain Comparison: We further analyze the spatial–frequency characteristics of the restoration results using the band-averaged 2-D log power spectra shown in Fig. 14. In particular, the 2-D log power spectrum is computed for each spectral band and then averaged across all bands. We present a frequency-domain analysis using *KAIST_scene04_patch_008* (first and second rows of Fig. 14) as a representative example, with similar observations across the other samples. SMDS-Net and SERT exhibit

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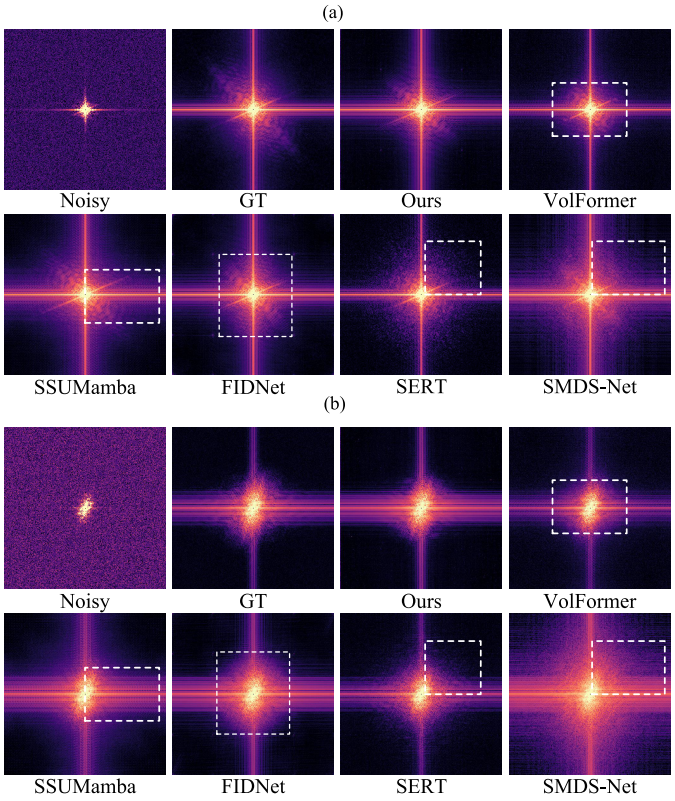


Fig. 14. Frequency-domain visualization of representative samples using band-averaged 2-D log power spectra under Case 1. (a) and (b) Correspond to *KAIST_scene04_patch_008* and *peppers_0503-1315_patch_34*, respectively.

scattered bright speckles in the off-center high-frequency regions, indicating residual noise. VolFormer, despite a relatively clean background, loses the off-axis diagonal components, suggesting that directional details are mistakenly treated as high-frequency outliers. As for SSUMamba, it introduces noticeable energy spreading in the Fourier domain along the horizontal and vertical axes, where the disordered energy distribution may indicate residual structured artifacts. FIDNet exhibits an expanded central region in the frequency spectrum, implying overemphasis on low-frequency signals. On the contrary, Diff-DUN produces spectra that are energetically aligned with the GT, effectively suppressing high-frequency noise while preserving directional textures with minimal artifacts. To complement the frequency-domain analysis, we further assess edge preservation using absolute gradient-error maps. As shown in Fig. 15, Diff-DUN produces fewer residuals around edges and textured regions. These results suggest that the improvement of Diff-DUN extends beyond global noise suppression to the preservation of spatial–frequency structures and edge details.

C. Experiment on Real HSIs

To further validate the generalization capability, we evaluate all methods on the real-world Australia[5] and Urban datasets.[6] Since no clean reference is available, we assess the denoising performance based on visual quality, with parameters of all methods carefully tuned with the official settings.

Fig. 16 shows the HSIs recovered by different methods. As can be seen, our model removes complex noise while maintaining high fidelity in texture details, achieving competitive visual quality.

[5]http://remote-sensing.nci.org.au/u39/public/html/index.shtml
[6]http://www.tec.army.mil/hypercube/

TABLE III
ABLATION OF THE TT BLOCK UNDER CASE 4. RESULTS ARE AVERAGED OVER 562 TEST SAMPLES. "W/O" DENOTES "WITHOUT"

Method	Performance			Complexity
	PSNR (dB) ↑	SSIM ↑	ERGAS ↓	Params (M)
Full TT block	39.00	0.9439	48.47	7.88
Spatial–Spectral Only	38.48	0.9386	51.27	2.55
w/o LDM Prior	38.81	0.9391	49.46	5.56
w/o Spatial Flow	36.88	0.8929	300.45	4.52
w/o Spectral Flow	38.58	0.9386	52.45	6.75
w/o Frequency Flow	38.51	0.9388	51.26	4.87
w/o SFI Module	37.83	0.9355	55.13	5.67
w/o MHCA Module	38.77	0.9420	49.75	7.38

D. Effect of the Degradation-Free Prior

In Diff-DUN, the degradation-free prior is a learned latent representation rather than a directly visualizable image. It encodes restoration-relevant spatial–spectral characteristics, including structural layouts, textural responses, and edge information. In particular, the prior $Z_0 \in \mathbb{R}^{N\times C_z}$ contains $N = 16$ latent tokens, each with a feature dimension of $C_z = 256$.

Given that the latent prior is not directly interpretable in the image domain, we evaluate its contribution through the restored outputs and their corresponding absolute error maps. As illustrated in Fig. 17, compared with the variant without the LDM prior, the full model preserves clearer textural details and sharper edge structures while producing weaker error responses around object boundaries. These observations indicate that the generated prior provides complementary spatial–spectral guidance for faithful HSI restoration.

E. Ablation Studies and Model Analysis

We conduct ablation studies to assess the contributions of the principal architectural components and the effects of key model settings. We further analyze the computational efficiency and training stability of Diff-DUN. Unless otherwise specified, experiments use the three-stage Diff-DUN.

1) Effectiveness of the Four Flows: To verify this design empirically and further examine the potential redundancy among the four flows, we have conducted a systematic ablation study. The ablation results in Table III lead to several observations. Removing the spatial flow induces the most severe performance degradation, with ERGAS increasing from 48.47 to 300.45. This indicates that reliable local structure modeling is crucial to HSI denoising. The spectral flow preserves spectral correlations, while the frequency flow suppresses frequency-domain artifacts that are less explicit in the pixel domain. Their joint effect explains why the full TT block consistently outperforms the spatial–spectral-only variant. Regarding the LDM prior flow, it offers a different form of guidance. It complements the information flows

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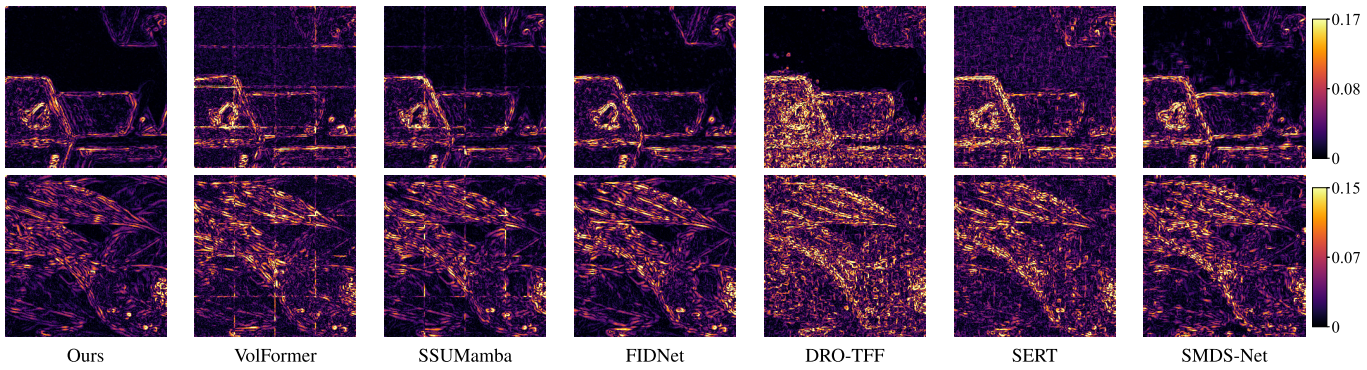


Fig. 15. Gradient-magnitude error maps of representative denoising results on Band 14 under Case 1. From top to bottom, the rows correspond to *KAIST_scene04_patch_008* and *peppers_0503-1315_patch_34*, respectively. Brighter regions indicate larger gradient residuals with respect to the GT, while darker regions indicate smaller residuals.

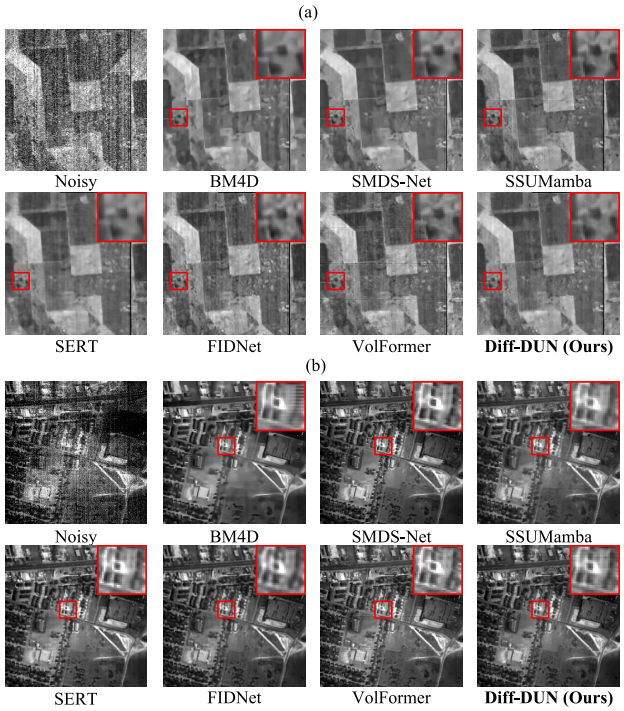


Fig. 16. Visual comparison of denoising results on real-world HSI datasets. (a) Australia dataset, band 51. (b) Urban dataset, band 109.

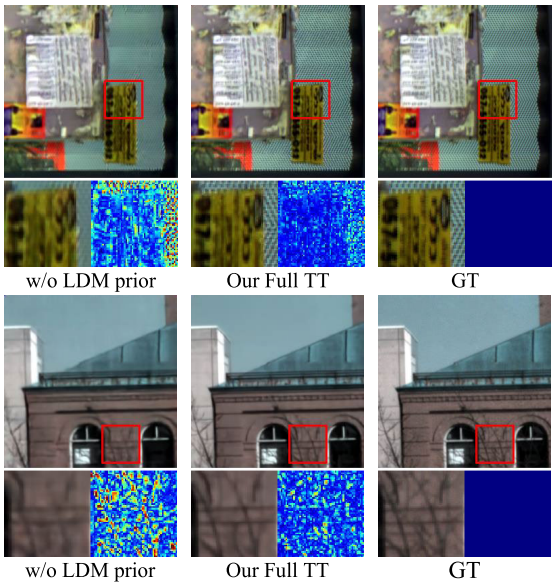


Fig. 17. Visual comparison of denoising results with and without the LDM prior on two representative samples: *rsh_0406-1443_patch_27* (R:26, G:18, and B:9) from the ICVL dataset and *imga1_patch_03* (R:27, G:21, and B:20) from the Harvard dataset under Case 2.

derived from the observed HSIs from a generative perspective, yet its individual PSNR gain is modest. Although its absence does not lead to catastrophic failure, it still provides additional improvement when working alongside the other flows. This suggests that the prior information serves as a beneficial supplement to the other flows. It is worth noting that removing the SFI module reduces PSNR to 37.83 dB, confirming the benefit of cross-flow information exchange. In addition, replacing MHCA with simple projection fusion also leads to a measurable performance decrease, suggesting that attention-based integration is crucial for aggregating features from different flows.

2) Computational Efficiency: Table IV presents a comprehensive comparison of different methods in terms of restoration quality, FLOPs, average inference time, and peak GPU memory consumption. As evident from Table IV and Fig. 1, SMDS-Net, VolFormer, and DRO-TFF exhibit substantially higher computational costs than other methods.

TABLE IV

COMPARISON OF RUNTIME EFFICIENCY, GPU MEMORY, AND RESTORATION QUALITY UNDER CASE 1 OVER 562 TEST SAMPLES. THE REPORTED TIME IS THE AVERAGE RESTORATION TIME PER HSI

Method	Quality		Computational Cost		
	PSNR (dB)	SSIM	FLOPs (G)	Avg. Time (s)	GPU Mem. (MB)
BM4D (CPU) [5]	31.9948	0.8117	N/A	36.3023	N/A
NGMeet (CPU) [50]	36.5215	0.9200	N/A	49.6618	N/A
SMDS-Net (GPU) [15]	36.2449	0.9179	24996.86	55.7786	1635.06
FIDNet (GPU) [51]	36.6416	0.9227	73.37	0.0657	1893.06
SERT (GPU) [53]	36.4502	0.9133	268.94	0.0654	526.61
VolFormer (GPU) [52]	36.3591	0.9105	25124.89	1.6541	2718.50
SSUMamba (GPU) [54]	37.7040	0.9302	187.16	0.2160	407.35
DRO-TFF (GPU) [7]	34.4906	0.8714	31039.03	18.8780	246.42
Diff-DUN (GPU)	**38.5313**	**0.9392**	162.52	0.0782	580.00

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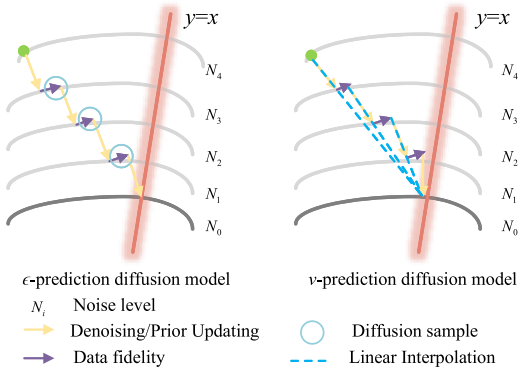


Fig. 18. Iterative process of v-prediction and ϵ-prediction.

TABLE V

ABLATION STUDY OF DIFFERENT DIFFUSION PREDICTION PARAMETERIZATIONS UNDER CASE 1 AND CASE 3. PERFORMANCE VALUES ARE AVERAGED OVER 562 TESTING SAMPLES

Case	Prediction	PSNR ↑	SSIM ↑	ERGAS ↓
Case 1	ϵ-prediction	38.44 ± 2.44	0.9387 ± 0.031	51.63 ± 16.88
	v-prediction	38.53 ± 2.47	0.9392 ± 0.031	51.14 ± 16.74
Case 3	ϵ-prediction	42.55 ± 2.51	0.9710 ± 0.017	31.82 ± 10.47
	v-prediction	43.27 ± 2.58	0.9748 ± 0.015	29.29 ± 9.82

In particular, the high computational costs of SMDS-Net, DRO-TFF, and VolFormer mainly arise from HySime subspace estimation and cube aggregation, image-specific optimization over 4000 iterations, and volumetric self-attention, respectively. On the contrary, Diff-DUN achieves a competitive tradeoff between reconstruction accuracy and computational efficiency.

3) Performance and Stability of Two Prediction Parameterizations: To ensure stable training of the LDM, we compare two prediction parameterizations [59]. We first adopt the standard ϵ-prediction, but observe that the loss is less numerically stable during later training. To mitigate this, we switch to v-prediction [43], which predicts the velocity vector $v_t = (\bar{\alpha}_t)^{1/2}\epsilon - (1-\bar{\alpha}_t)^{1/2}Z_0$ following the formulation in Section III. Empirically, v-prediction exhibits superior numerical stability and convergence compared to ϵ-prediction in Diff-DUN. As illustrated in Fig. 18, v-prediction may have the ability to effectively linearize the probability flow, leading to straighter sampling trajectories. This property effectively avoids the numerical difficulties that commonly arise when the prediction target changes dramatically in magnitude with the noise level [60]. Furthermore, we compare the performance of v-prediction against ϵ-prediction. As shown in Table V, the two variants achieve comparable performance under Case 1, while v-prediction brings clear improvements under Case 3, improving PSNR from 42.55 to 43.27 dB. These results confirm that v-prediction is more suitable for our prior generation, particularly under the band-dependent noise setting.

4) Effect of the Number of Unfolding Stages: The quantitative results in Table VI demonstrate that the benefit of the LDM prior becomes more pronounced as the number of unfolding stages increases. It is noted that the five- and ten-stage unfolding networks adopt the "parameter-sharing" strategy, where the parameters from the 2nd to the $(K-1)$th stages are shared. This strategy allows the network depth to be increased while avoiding a proportional increase in trainable parameters. As reported in Table VI, the performance gap gradually enlarges with deeper unfolding. This suggests that the diffusion prior offers complementary degradation-free guidance across the iterative process.

TABLE VI

ABLATION STUDY OF THE LDM PRIOR WITH DIFFERENT NUMBERS OF UNFOLDING STAGES UNDER CASE 4

Method	PSNR (dB)		
	3 stages	5 stages	10 stages
Full Model	39.00	40.01	40.57
w/o LDM	38.81	39.69	40.12
Difference	+0.19	+0.32	+0.45

TABLE VII

STAGEWISE PARAMETERS USED IN THE THREE-STAGE UNFOLDING MODULE UNDER DIFFERENT NOISE CASES

Case	ρ	$\boldsymbol{\tau}_{1:3}$	$\boldsymbol{\sigma}_{1:3}$
Case 1	1.000	[0.0438, 0.0389, 0.0415]	[0.2092, 0.1973, 0.2038]
Case 2	1.000	[0.0180, 0.0170, 0.0184]	[0.1340, 0.1304, 0.1358]
Case 3	1.000	[0.0181, 0.0171, 0.0184]	[0.1347, 0.1309, 0.1356]
Case 4	1.000	[0.0905, 0.0821, 0.0853]	[0.3008, 0.2866, 0.2921]

TABLE VIII

ABLATION STUDY OF THE NUMBER OF DIFFUSION STEPS UNDER CASE 3

Diffusion Step	0	2	4	8	16
PSNR (dB)	42.61	42.98	43.17	43.25	**43.27**

5) Parameter Analysis: For the unfolding module, the parameter ρ is kept constant across all stages, whereas τ_k is learned independently for each stage. The corresponding denoising strength at the kth stage is defined by $\sigma_k = (\tau_k/(\rho+\varepsilon))^{1/2}$, where ε is a small constant to prevent numerical instability. For the three-stage model, the fixed penalty parameter and the learned stage-dependent parameters under different noise cases are reported in Table VII.

6) Effect of the Number of Diffusion Steps: To investigate the impact of the number of diffusion steps T, we evaluate Diff-DUN with varying T. As shown in Table VIII, at $T=0$, reverse diffusion is disabled, and the prior is directly extracted from the degraded HSI to guide restoration, yielding the lowest performance. This confirms that progressive latent refinement via reverse diffusion is beneficial. As T increases, performance improves at the cost of more computation. Considering both accuracy and efficiency, we set $T=16$ in all experiments.

7) Empirical Convergence and Training Stability: Classical HQS optimization with explicit handcrafted regularizers can admit convergence guarantees under suitable assumptions. However, Diff-DUN replaces the handcrafted prior with a learnable denoiser conditioned on a stochastic diffusion prior, rendering theoretical analysis a largely unexplored area. We therefore empirically assess optimization stability. As shown in Figs. 19 and 20, training losses in both phases decrease and gradually stabilize for Cases 1 and 4. These observations empirically confirm stable training under the evaluated i.i.d. and band-dependent non-i.i.d. noise settings.

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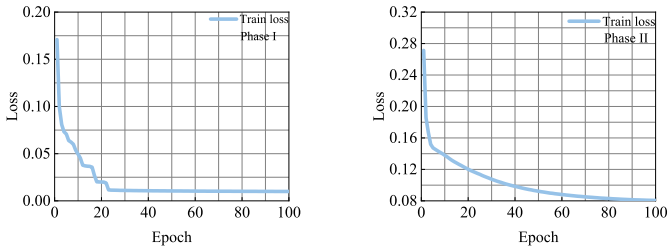


Fig. 19. Training loss curves for Phases I and II under Case 1.

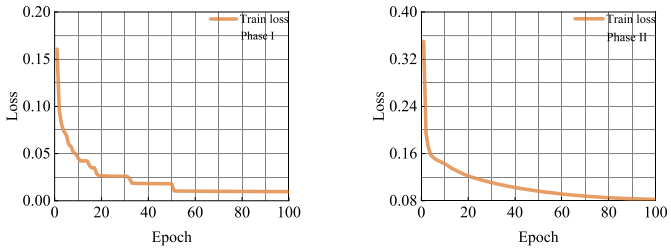


Fig. 20. Training loss curves for Phases I and II under Case 4.

V. Conclusion

In this article, we propose a latent diffusion prior-enhanced frequency-aware DUN for HSI denoising. Diff-DUN integrates an HQS-based unfolding network with a conditional LDM, striking a balance between observation consistency and perceptual quality. In particular, we employ a two-phase strategy to learn a degradation-free prior from clean HSIs and generate it from noisy observations. Within each unfolding stage, we propose a frequency-aware TT block to progressively integrate degradation-free prior flow, spatial, spectral, and frequency flows. The spatial and spectral flows characterize spatial–spectral features and adaptively select observation-consistent information from the prior, while the frequency flow models the distinct responses of structural details and noise to improve edge preservation and noise suppression. Through these interactions, the TT block bridges the LDM prior and HQS updates for faithful HSI denoising. In addition, Diff-DUN employs stage-dependent denoising strengths and a more suitable v-prediction strategy for prior generation. Extensive experiments validate the effectiveness and efficiency of Diff-DUN.

Although Diff-DUN operates with only 16 diffusion steps, the iterative prior sampling and two-phase training still impose considerable computational demands. Reducing the number of network evaluations via few-step or one-step diffusion is thus a primary direction for future optimization. In addition, the framework is formulated for Gaussian noise, consistent with our problem setting. However, real-world HSIs often suffer from more complex degradations, including impulse, stripe, and deadline noise. Extending our method to handle such mixed cases is therefore a promising future direction.

Acknowledgment

The authors are sincerely grateful to the editors and the anonymous reviewers for their insightful and constructive comments, which have greatly improved the quality of this article. They also thank the authors of [29] for their generous assistance.

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